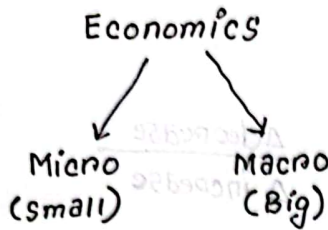


The Economic Terms of ECO101

1. Scarcity - Limited



Macro: Deals with performance, structure, behaviour and decision making of an economy as a whole.

Micro: Deals with study of choices that individuals like you & me and business make regarding the allocation of resources.

2. Factors of Production (FOP)

- Land
- Labour
- Capital
 - financial capital
 - Physical capital
- Entrepreneurship

- Land earns rent
- Labour earns wages
- Capital earns interest
- Entre. earns profit

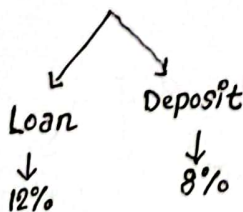
3. Social Interest & Self-Interest.

Taxation

4. Trade off - Opportunity cost (OC) * → giving up one thing to get something else.
(Give and Take)

✓ Marginality / Marginal Analysis:

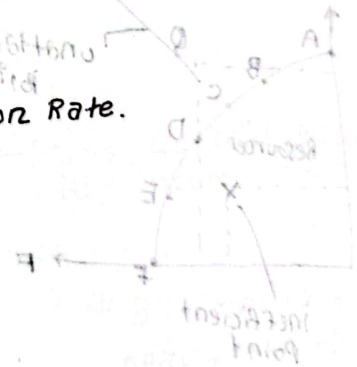
6. Interest rate



$$\text{Profit} = (12 - 8)\% = 4\%$$

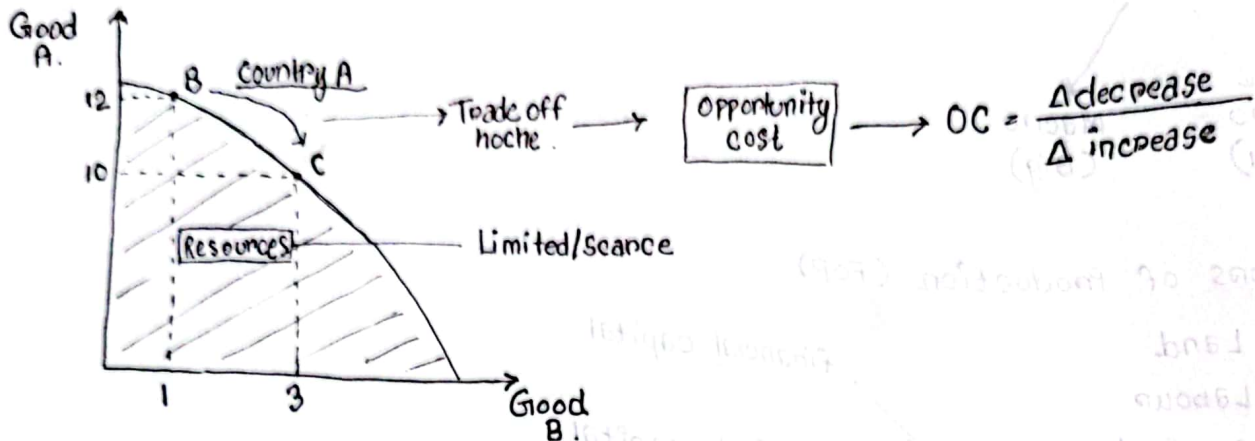
Real Interest Rate

% Int Rate - Inflation Rate.



Models of Economics

1. Production Possibility Curve/Frontier : PPC/PPF



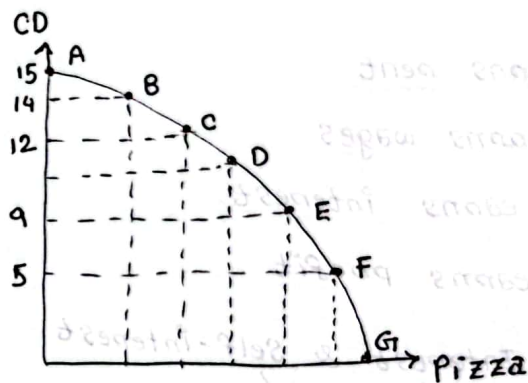
Problems

* Aste Aste

give it in any axes.

Graph (a)

Points	Pizza	CD	OC
A	0	15	$\frac{15-14}{1-0} = 1$
B	1	14	$\frac{14-12}{2-1} = 2$
C	2	12	$\frac{12-9}{3-2} = 3$
D	3	9	
E	4	5	
F	5	0	

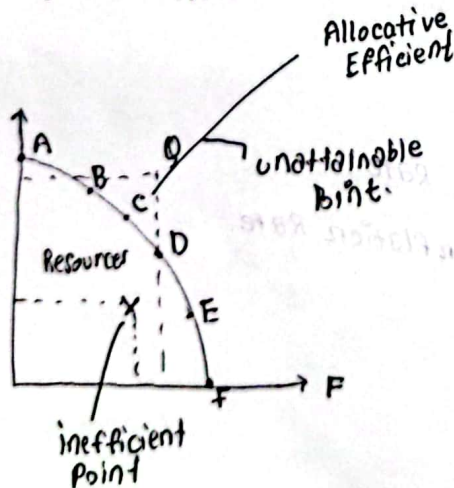


(b) Find the OC of Pizza.

$\Rightarrow OC = \underline{\hspace{2cm}}$

* Increasing OC as we produce more of one Good.

(Sacrifices)

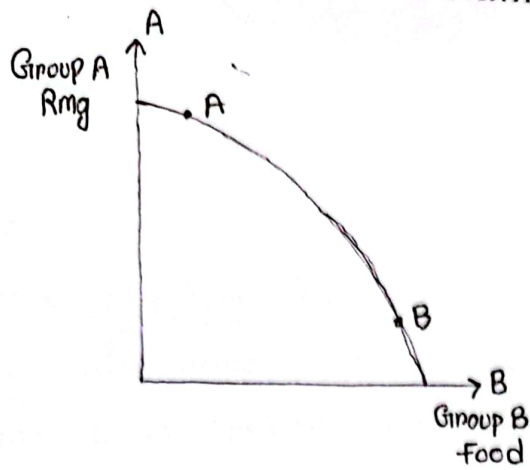


+ New Tech
+ Human Capital
Econ. growth

Why OC always keep increasing?

because we cannot move our resources from one sector to another overnight.
it will in cause increase in sacrifice as well OC.

#

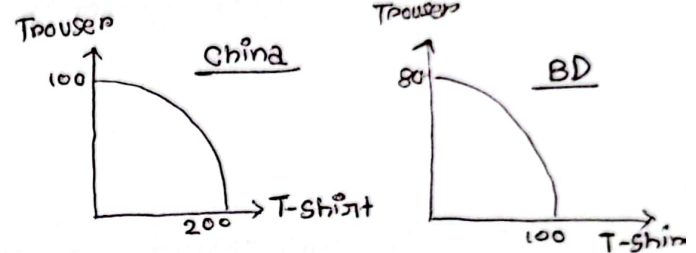


'To start a Trade relationship with other nations'

Socialism - divide for all

Law of Absolute and Comparative Advantage:

	Trousers	T-Shirts
China	100	200
BD	80	100



comparative Advantages: When a country has a lower OC in producing something then that country should produce it. But when a country has a higher OC in producing something then it should import it from other countries.

Calculating OC = $\frac{\Delta \text{Decrease}}{\Delta \text{Increase}}$ → decreasing production → Give up
→ Produce

	Trousers	T-shirt
China	$\frac{200}{100} = 2.00$	$\frac{100}{200} = 0.500$
BD	$\frac{100}{80} = 1.25$	$\frac{80}{100} = 0.8$

import karte. ← (for China's T-shirt and BD's Trousers)

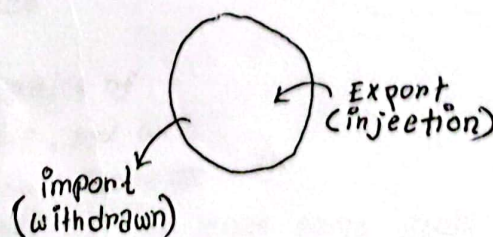
• Higher OP jar tara import karte and trade lower OP jar receive karte.

Rule → will last until 2026. as when we will become middle income country.

↓
Everything but Arms

• Tariff - Tax on Trade.

BD import karte, export karte.



Galaxy A52

* [BD should start entering trade blocks].

Demand & Supply

Demand:

• Competitive / Market:

An arrangement that enables buyers and sellers to get information and to do business with each other.

• Diff types of Markets:

- i. Market of goods . (Unimant, Jamuna) ✓ mainly study .
- ii. Market of Services . (Banks)
- iii. Money market (Stock, bonds)

• Competitive market: A market that has many buyers and less sellers .

• Money price & Relative Price

money is the biggest factor of O.P.

• Law of Demand: i. $P \downarrow \rightarrow Q_d \uparrow$

↓
Inverse relationship

between price & Quant
demanded (Q_d) .

ii. $P \uparrow \rightarrow Q_d \downarrow$

Proven Substitution Effect

↓
tells us where the price of some
good rises, the others remain same,
its relative price - OC rises .

$P \uparrow \rightarrow Q_d \uparrow$
Coke Pepsi

Imp:

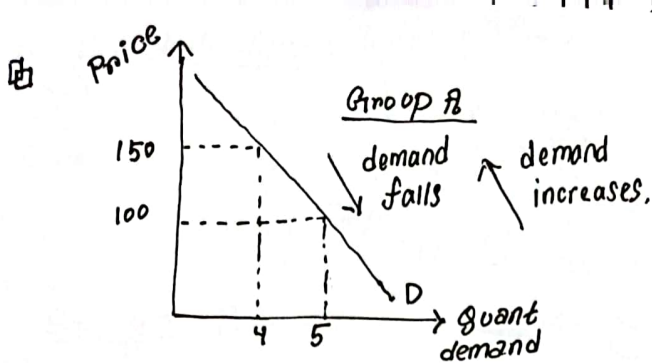
$P_{Coke} \uparrow \rightarrow \text{Income (constant)} \rightarrow Q_d \downarrow$

↓
Income Effect

↓
tells us where price of
some good rises, ppl with
unchanged income decrease the
quantities demanded of at least some goods & services .

• Two goods - at some situation
(Coke vs Pepsi)

$P \uparrow \rightarrow Q_d \uparrow$
Coke Pepsi
 $P \uparrow \rightarrow Q_{Coke} \downarrow$



Shift - When should we shift?

Whenever (price)
related info,
for eg: Price of
sugar has increased,
from ... to

• You don't have to shift (0.01%)
comes.

← decreasing
(left)

→ increasing
(right)

Shifts when change in demand

for eg: During Ramadan,
Price of sugar
increases (99.99%)
comes.

6 demands of changes in demand

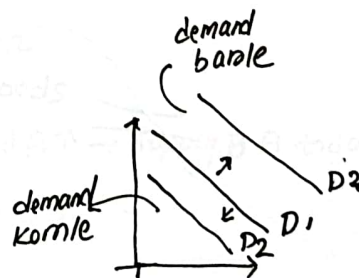
Substitute goods - goods which gives
same level of satisfaction

Complement goods - good which
complements the other
goods -
eg - Tea & Sugar.

- i. Prices of related goods.
- ii. Expected future prices.
- iii. Income $\rightarrow$ Income $\uparrow \rightarrow Q_d \rightarrow$ without knowing
what kind of good
we are dealing with
- we can't get
down to conclusion.
- iv. Expected future income
- v. Population $\uparrow - D \uparrow$
- vi. Preferences $\uparrow - D \uparrow$

$P_{\text{coke}} \uparrow \rightarrow Q_{\text{Pepsi}} \uparrow - D \uparrow$ Demand
increase.

$P_{\text{ink}} \downarrow \rightarrow Q_{\text{Printer}} \uparrow - D \uparrow$ Demand
increase.



$P_{\text{sugar}} \uparrow \rightarrow Q_d \uparrow =$ demand
increase.

Income $\uparrow \rightarrow Q_d \uparrow$ - If normal goods - $D \uparrow$ (Demand rise).

Income $\uparrow \rightarrow Q_d \uparrow$ (Inferior good) $\rightarrow D \downarrow$ (Demand falls)

[Public transport]
for eg.

Galaxy A52

IV. Expected future income: $I \uparrow \rightarrow D \downarrow$ - Demand decrease.

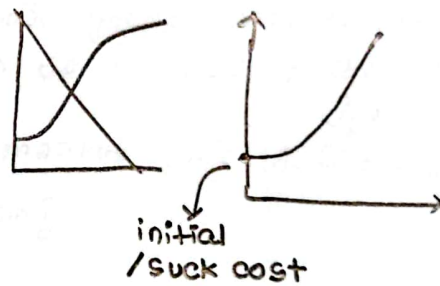
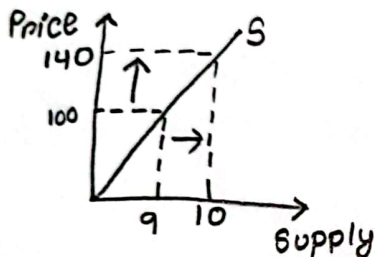
2.2

Q Low of Demand: $P \uparrow \rightarrow Q_d \downarrow$ (substitute effect)
 $P \downarrow \rightarrow Q_d \uparrow$ (income effect).
 : Price of Q_d are 'inversely' related.

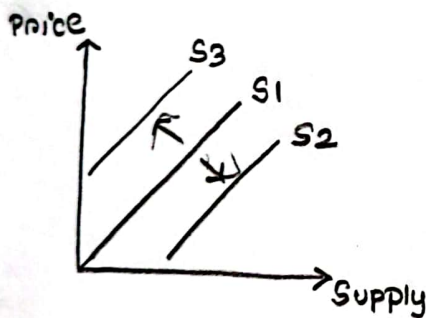
Supply: Producer. demand Consumer.

Law of Supply: $P \uparrow \rightarrow Q_s \uparrow$
 $P \downarrow \rightarrow Q_s \downarrow$

Supply Curve



Anything in
eco, right
shift, mane
bana. and left
shift mane koma



Change in supply.

- i. Prices of Factors of Production
- ii. * Prices of related produced goods
- iii. * Expected future prices $\rightarrow$ Good A $\uparrow \rightarrow$ Supply B A today $\downarrow$
- iv. Numbers of Suppliers $\downarrow \rightarrow S \downarrow$
- v. Technology $\uparrow \rightarrow S \uparrow$
- vi. The State of nature $\uparrow \rightarrow S \uparrow$

Substitute good: such a good that they can produce using the same ingredient.

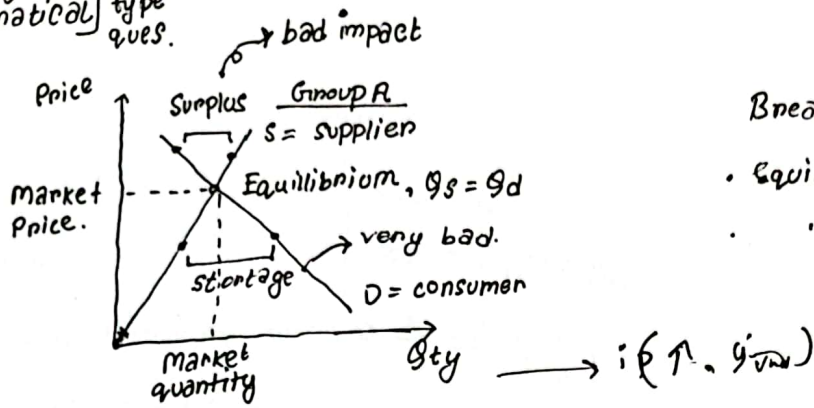
$P_{\text{Coke}} \uparrow \rightarrow Q_{\text{supply Pepsi}} \downarrow$ (decrease)

Complement good: Bi-product, when you're producing something anekta kichu produce hoye jawa.
 for eg ink theke pen hoilo.

$P_{\text{wood in furn}} \uparrow \rightarrow Q_{\text{boards}} \uparrow$

Topic 1: PPF & (AA, CA) Examples of Demand & supply

- Shifting
- Mathematical



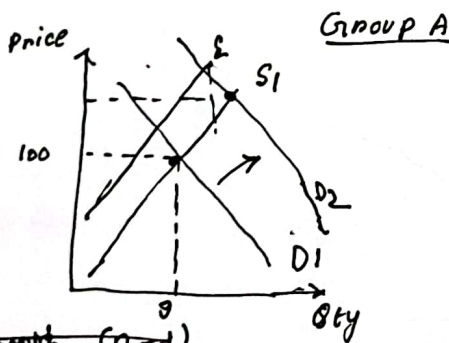
Breaking of equilibrium is bad.

- Equilibrium en upare - Surplus
- " " niche - Shortage

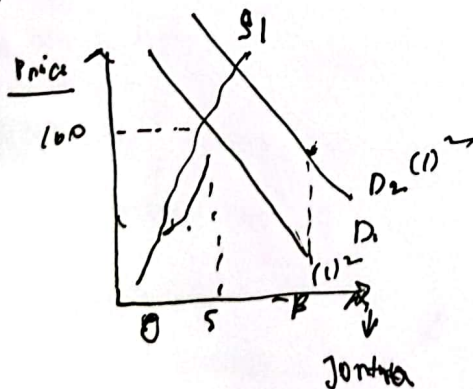
Q. Diagrammatical

Suppose, due to rise in the income level, people's reference for group A has changed. (Note: Group A is ~~not~~ normal good). Further, due to the recent rise of labor wage production of Group A is obstructed.

What will happen to Group-A market price & (Qty)? Illustrate your answer with a app. diagram? (PT) (uncertain)



Support (1-1)
4. question is same. but ~~in~~ in ques, there will be a fall in ~~price~~ ~~quantity~~.
7b.

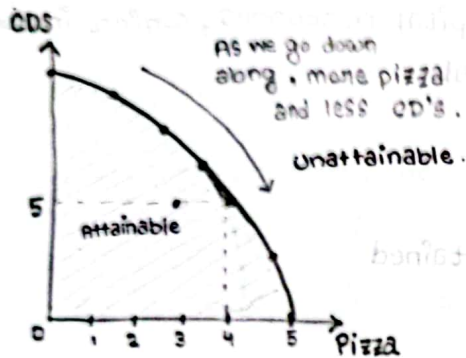


5. Same ques but ~~input~~ input of labour wage is ~~higher~~ higher.

EconomicsPractice - Quiz 1

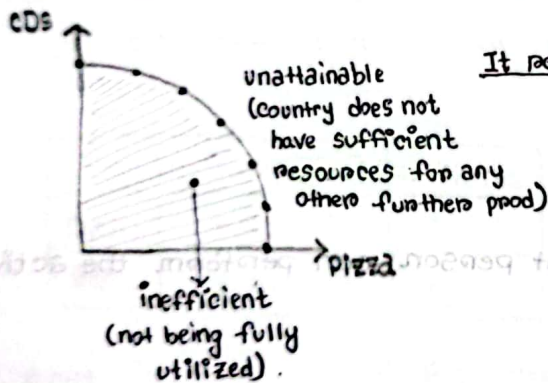
formula of opp cost = $\frac{\Delta \text{decrease A}}{\Delta \text{increase B}}$

Opportunity cost increases as we move along PPF curve.



Points	Pizza	CDs	OC of Producing Pizza
A	0	15	
B	1	14	$\frac{(15-14)}{(1-0)} = 1$
C	2	12	$\frac{(14-12)}{(2-1)} = 2$
D	3	9	$\frac{(12-9)}{(3-2)} = 3$
E	4	5	$\frac{(9-5)}{(4-3)} = 4$
F	5	0	$\frac{(5-0)}{(5-4)} = 5$

Production Possibility frontiers (PPF)



It represents:

Scarcity

Production efficiency

Trade-off

Opp. Cost (OC)

occurs

- if we cannot produce more of 1 good without producing less of some other good.

• why does the PPF bowed outward?

- It is due to the law of increasing opportunity cost and because resources are not all equally productive in all activities.

• why OC always keeps increasing?

- This is because when we move production from one point to another OC tends to increase.

• Allocative efficiency: When we cannot produce more of any good without giving up some other good that we value more highly.

☞ Economic Growth: The expansion of production possibilities - and increase in the standard of living -

- factors
 - Technological change: dev of new goods and of better ways of producing goods & services.
 - Capital accumulation: growth of capital resources, which includes human capital.

☞ The production possibilities frontier represents
- the maximum levels of production that can be attained.

☞ The bulging-out of the PPF curve represents
- increasing opportunity cost.

☞ Trade → Exchange of goods.

- A person has a comparative Advantage if that person can perform the activity at a lower OC than anyone else.

☞ Acc to Comparative advantage, countries gain trade because
- world trade can rise when each country specializes in what it does relatively best.

Calculation

	olive	olive oil
Spain	60	11
Greece	8	5

$$\frac{11}{60} = 0.2 \quad \left| \quad \frac{60}{11} = 5.5 \right.$$

$$\frac{5}{8} = 0.6 \quad \left| \quad \frac{8}{5} = 1.6 \right.$$

$$OC = \frac{\Delta \text{decrease}}{\Delta \text{increase}}$$

→ OC (0.2, 5.5)

→ OC (0.6, 1.6)

Spain sacrifices 0.2 Olive oil for every olive.

and Greece " ~~only~~ ^{0.6} " " " " " "

Q = which country has a comp advantage in producing olives.

= Spain

Q. in olive oil, = Greece

Q Absolute Advantage refers to the ability of to produce more or better goods and services than somebody else.

• Comparative advantage refers to the ability of to produce goods at a lower OC, not necessarily at a greater quantity.

Q:

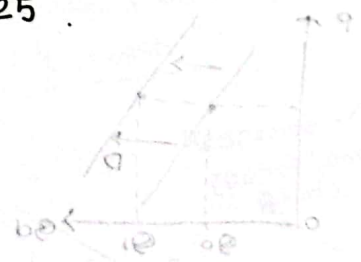
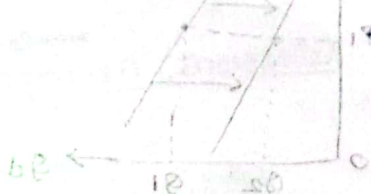
	Bangladesh	China
Corn	9 acres	8
Rice	3 acres	2

Q. what is the OC of BD for producing 100 bushels of corn?

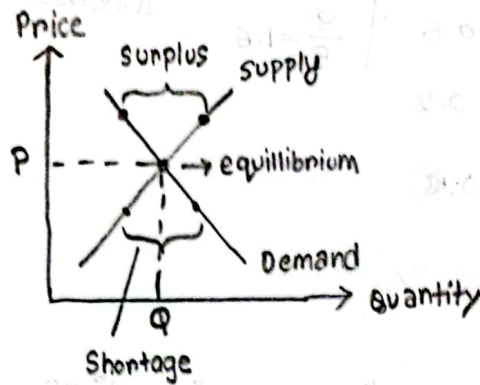
$$OC = \frac{\text{corn}}{\text{rice}} = \frac{9}{3} = 3 = 3 \times 100 = 300$$

Q. what is the OC of china for producing 100 bushels of rice?

$$OC = \frac{2}{8} = 0.25 = 0.25 \times 100 = 25$$



▣ Demand and Supply



In short, demand refers to the curve and quantity demanded refers to a specific point on the curve.

Law of Demand

when the price goes up $\uparrow$ → the quantity demanded goes down $\downarrow$.

$$P \uparrow \rightarrow Q_d \downarrow$$

and if, when the price goes down $\downarrow$ → the quantity demanded goes up $\uparrow$.

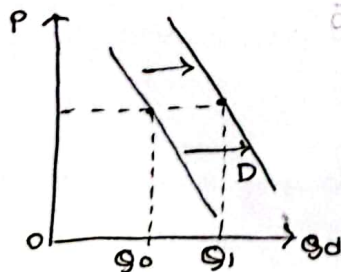
$$P \downarrow \rightarrow Q_d \uparrow$$



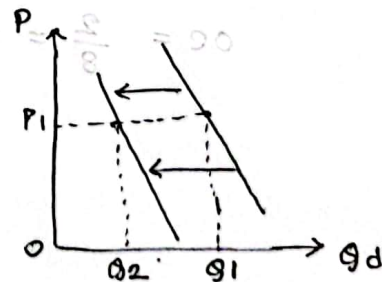
→ Shift in demand occurs when any other factors occur except than price

• When price changes, Quantity demand changes.

For normal goods - whose demand increases with an increase in income.



Higher income → higher demand

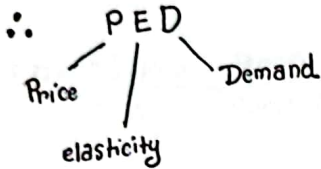
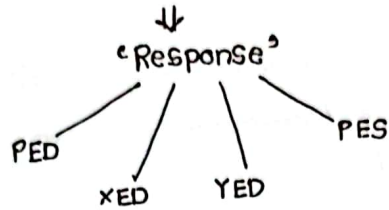


Lower income → lower demand

In inferior goods, it's the opposite; higher → Lower
Lower → higher.

Elasticity

- Measure of a variable's sensitivity to a change in another variable



$$PED = \frac{\% \Delta D}{\% \Delta P} \quad \left(\% \Delta = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100 \text{ (End-point formula)} \right) - X$$

↓
Instead we will use, Mid-point formula → $\% \Delta = \frac{\text{New} - \text{Old}}{(\text{New} + \text{Old})/2} \times 100$
(more precise).

Calculation

	Old	New
Price	3.10	3
Qty	9	10

4 values
must

$$PED = \frac{10 - 9}{(3 + 3.10)/2} \times 100 = \frac{1}{3.05} \times 100 = -3.27$$

$$\text{or, } = \frac{9/10}{(3.10 - 3)/2} \times 100 =$$

$$PED = -3.21$$

↓
always comes
negative.

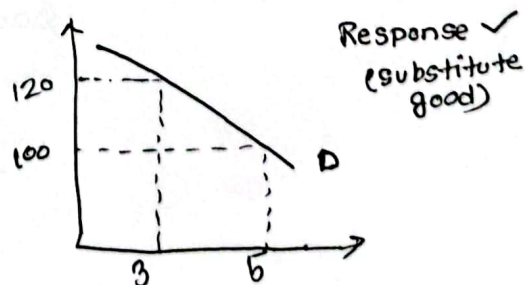
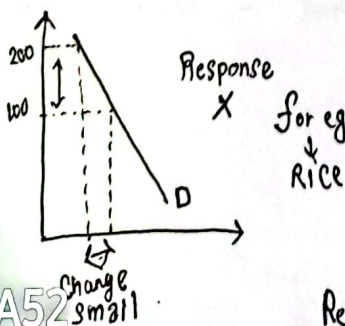
- ① $PED > 1$: Elastic Good. 
- ② $0 < PED < 1$: Inelastic Good. 

- ③ $PED = 0$ (Perfectly inelastic good)



- ④ $PED = \infty$ (Perfectly elastic good) = 

Good A (Inelastic)



$$\text{Revenue} = P \times Q, \text{ Profit} = \text{Revenue} - \text{cost}$$

Imp:
XED

Topic 1 : PPF & (AA, CA)
Response

Eco101

Xenoss ~~Elasticity~~ Demand

What will happen to Good A's demand to a change in price of Good B.

Reminders

$PED > 1$ elastic

$0 < PED < 1$ inelastic

$PED = 0$, Perfectly inelastic

$PED = \infty$, " elastic

$$XED = \frac{\% \Delta Q_d - \text{Good A}}{\% \Delta P - \text{Good B}}$$

$$\% \Delta = \frac{\text{New} - \text{old}}{\frac{(\text{New} + \text{old})}{2}} \times 100 \quad \text{Midpoint formula}$$

$$XED = Q_d = \frac{250 - 150}{\frac{(250 + 150)}{2}} \times 100 = 50\%$$

$$\frac{11 - 9}{\frac{11 + 9}{2}} \times 100 = 20\%$$

$$XED = \frac{20}{50} = 0.4$$

tea - 9 - 11

coffee - 150 - 250

Ans: $XED = 0.4$
(Substitute goods).

YED

Y_{income} ~~Elasticity~~ Demand
Response

+ve → normal good → $1 \uparrow$ → Income elastic good.
-ve → inferior good → $1 \downarrow$ → Income inelastic good.

$$YED = \frac{\frac{11 - 9}{\frac{11 + 9}{2}} \times 100}{\frac{60 - 50}{\frac{60 + 50}{2}} \times 100} = \frac{20}{18.2}$$

$$= \frac{20}{18.2}$$

$$= 1.098$$

$$= 1.10 \quad \text{Income elastic good.}$$

$$\frac{\Delta \% Q_d}{\% \text{Income}}$$

Galaxy A52

Topic 1 : PPF & (AA, CA)

Eco101

PES

P_{Price} E_{Elasticity} S_{Supply}

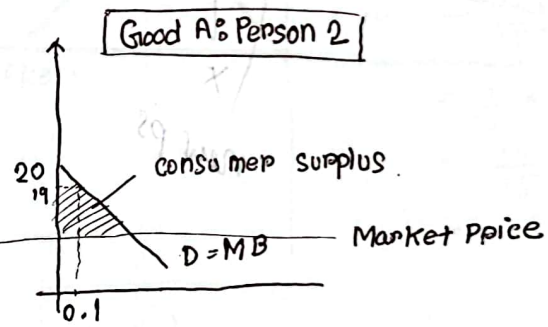
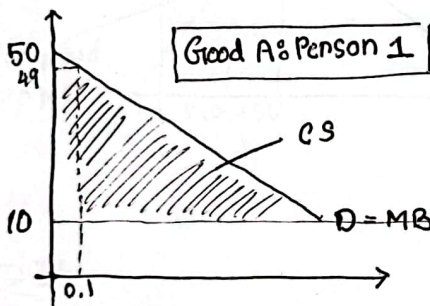
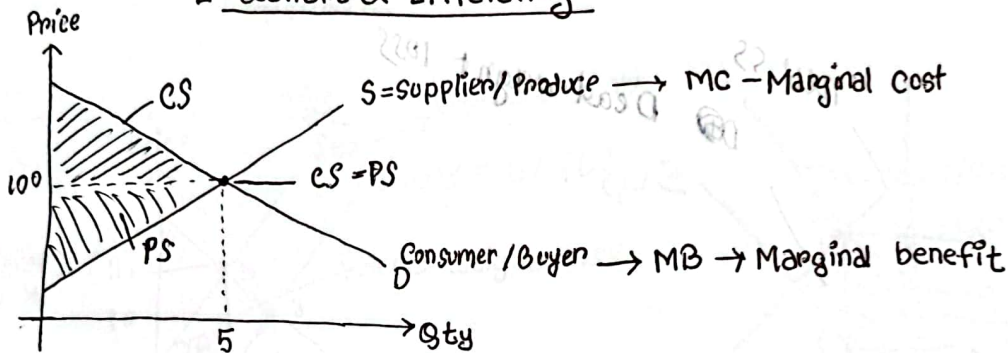
Table 4.1
from book
Economics 6th ed
Powell Matthews
(P&S)

PE D = (-ve) e ashe

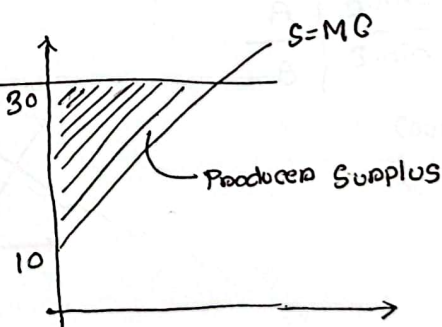
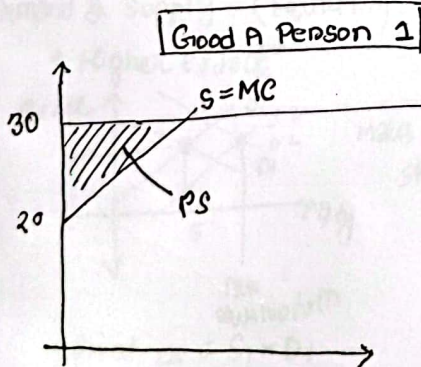
PES = (+ve) e ashe

$$\therefore PES = \frac{\% \Delta S}{\% \Delta P}$$

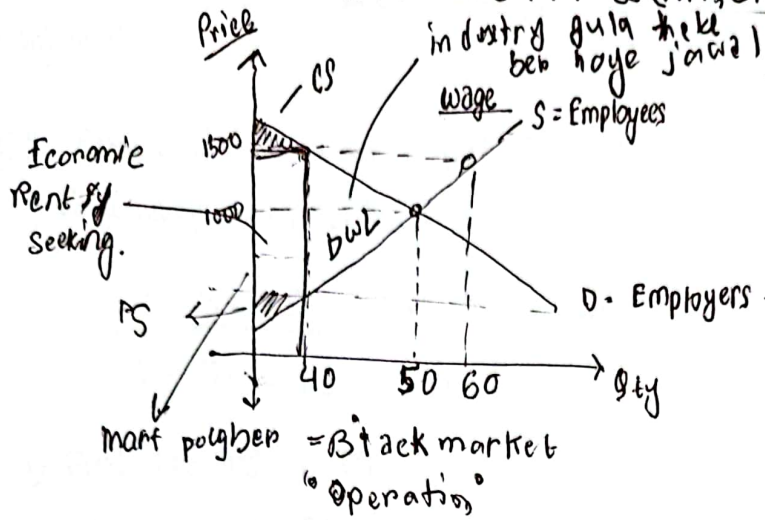
Welfare & Efficiency



Good A's market price = 10 tk (MB)



Topic 1: PPF & (AA, CA)

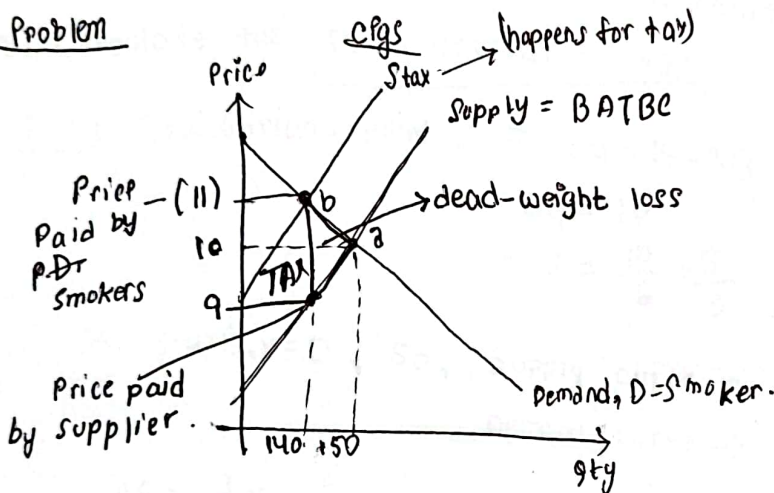


Eco101

Job market equilibrium to market

- ① Market distortion = labor - Labour simply.
- ② CS $\downarrow$
- ③ PS $\uparrow$
- ④ Economic rent sinking.
- ⑤ Dead-weight loss.

Problem



Tax = 2tk/cig.
Tax on suppliers =

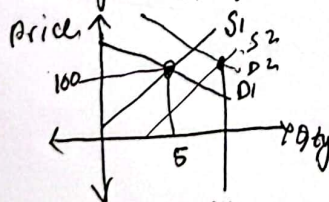
$$\begin{aligned} 150 & - 10 \\ 140 & - \frac{10}{150} \times 140 \\ & = \frac{1}{15} \times 14 \\ & = \frac{14}{15} \\ & = \end{aligned}$$

Economics:

① PPF/Comparative Advantage.

② Demand & Supply. (Equation)

* Higher effect*



$$OC = \frac{\Delta \text{De c (units)}}{\Delta \text{Price (units)}}$$



	T-shirt	Trouser
A	5 min (4/5)	4 min (5/4) = 20 c.
B	3 min	5 min

Count A = 5 min - 3 shirts
(30 min/1h)

$$\begin{array}{r} 12 \\ \times 3 \\ \hline 36 \end{array}$$

Create new table of units

③ Elasticity. (PED, XED, YED) Graphs.

response

$$PED = \frac{\Delta Q}{\Delta P}$$

$$YED = \frac{\Delta Q}{\Delta \text{income}}$$

④ Welfare & Efficiency.

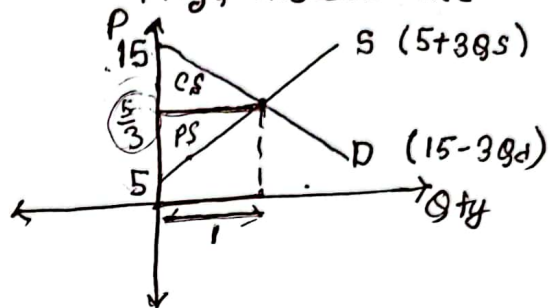
Galaxy A52

Problem

Q: Provided the below equations of Demand & Supply, answers the following questions:

$$P = 5 + 3Q_s \text{ - Supply curve}$$

$$P = 15 - 3Q_d \text{ - Demand "}$$



- 1) Find the CS & PS.
- 2) The gov decides to add 10Tk tax/unit. Find the new CS & PS
- 3) Calculate the DWL created by tax.

Ans: Equilibrium point, $5 + 3Q = 15 - 3Q$

$$6Q = 10$$

$$Q = \frac{10}{6} = \frac{5}{3}$$

at y-axis, $x = 0$, So, Supply curve $\rightarrow P = 5$

Demand curve $\rightarrow P = 15$



$$CS = \frac{1}{2} \times$$

base $\rightarrow$

